

LLMs, AI Tools & Google NotebookLM

Detailed Lecture Notes • Beginner to Intermediate • Study + Practical Use

Learning goal: By the end of this lecture, you should be able to explain what an LLM is, how modern AI tools work together, how to write useful prompts, and how Google NotebookLM can turn your own sources into a research and study assistant.

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1. Artificial Intelligence: The Big Picture

Artificial Intelligence (AI) is the field of computing concerned with building systems that can perform tasks that normally require human-like abilities such as understanding language, recognizing patterns, making predictions, solving problems, generating content and assisting with decisions.

AI is a broad umbrella. Machine learning is a major approach inside AI, deep learning is a major approach inside machine learning, and modern generative AI systems such as large language models are built using deep-learning methods.

A simple hierarchy

- Artificial Intelligence: the broad field.
- Machine Learning: systems learn patterns from data instead of relying only on hand-written rules.
- Deep Learning: machine learning using multi-layer neural networks.
- Generative AI: models that generate new text, images, audio, video, code and other content.
- LLMs: generative AI models specialized in working with language and language-like sequences.

Examples of AI applications

- Recommendation systems for videos, music and shopping.
- Speech recognition and voice assistants.
- Computer vision for images and video.
- Fraud detection and anomaly detection.
- Generative systems for text, images, code, audio and video.

2. Machine Learning and Deep Learning

Traditional software generally follows explicit instructions written by programmers. In machine learning, an algorithm learns statistical patterns from examples. During training, the model adjusts internal parameters so that its predictions become better according to an objective or loss function.

Training data → learning algorithm → trained model → new input → prediction/output

Neural networks

A neural network is a mathematical model made from connected computational units arranged in layers. Deep neural networks contain many layers and can learn increasingly complex representations. Transformer-based networks are especially important for modern language models.

3. What Is an LLM?

LLM stands for Large Language Model. It is a neural-network model trained on very large collections of text and other language-related data so that it can model relationships between tokens and generate useful sequences of language.

At a high level, an LLM repeatedly estimates what token is likely to come next given the previous context. Modern systems add many additional training and alignment techniques, so this simple description does not capture everything the system can do.

Common LLM abilities

- Answering questions and explaining concepts.
- Summarizing and transforming text.

- Translation and language practice.
- Writing and editing.
- Reasoning over provided information.
- Generating and explaining computer code.
- Extracting structured information from unstructured text.
- Using tools or external information when the surrounding AI system provides those capabilities.

Important point: an LLM is not a database in the ordinary sense. It stores learned statistical patterns in its parameters. When exact, current or source-grounded information matters, external retrieval or user-provided documents can be extremely useful.

4. How an LLM Processes a Prompt

A useful simplified pipeline is:

- Input: you provide a prompt or conversation.
- Tokenization: the text is divided into tokens, which may be words, parts of words, punctuation or other units.
- Context construction: relevant conversation and supplied information are placed into the model's context.
- Transformer processing: attention and neural-network layers transform the token representations.
- Next-token generation: the model assigns probabilities to possible next tokens.
- Decoding: a token is selected and the process repeats until the response is complete.

Tokens

A token is a unit used by the model's tokenizer. A token is not always equal to one English word. A long word may be split into several tokens, while some short words or symbols may be represented by single tokens. Tokenization affects context limits and usage costs in systems that charge by tokens.

Context window

The context window is the amount of information the model can consider within a particular interaction. A larger context can allow longer documents and conversations, but it does not automatically guarantee perfect attention to every detail.

Parameters

Parameters are learned numerical values inside a neural network. They are adjusted during training. The number of parameters is one indicator of model scale, but it is not by itself a complete measure of quality.

5. Training an LLM

Pre-training

During pre-training, a model learns broad language patterns from large datasets. A common objective is to predict missing or next tokens. This stage teaches the model general representations of language, facts present in the data, styles, patterns and relationships.

Fine-tuning

Fine-tuning trains a pretrained model further on a more focused dataset or objective. It can make a model more useful for a particular task, domain, style or instruction-following behavior.

Instruction tuning and alignment

Additional training methods can teach a model to follow instructions, be more helpful, and follow safety and behavioral requirements. Human or automated preference signals may be used as part of alignment pipelines.

Why can LLMs hallucinate?

A hallucination is an output that sounds plausible but is unsupported, incorrect or fabricated. Because language generation is based on learned patterns and probabilities, fluent wording is not proof that a claim is true. Reduce risk by providing authoritative sources, asking for evidence, checking important claims, and using retrieval or grounded workflows.

6. Prompt Engineering

Prompt engineering means designing instructions and context so an AI system is more likely to produce the intended result.

A strong prompt often contains five parts:

- Role/context: explain the situation.
- Task: say exactly what must be done.
- Inputs: provide the text, data or constraints.
- Output format: specify table, bullets, JSON, lesson, script, etc.
- Quality criteria: explain what a good answer must include or avoid.

Example

Weak: Explain AI.

Better: Explain AI to a first-year engineering student in simple English. Cover AI, ML, deep learning, generative AI and LLMs. Use one real-world example for each and finish with five exam points.

Useful prompt techniques

- Ask the model to use a specific audience and difficulty level.
- Give a source and ask it to stay grounded in that source.
- Request a structured output such as a table.
- Break a large task into stages.
- Ask for assumptions to be stated.
- For important facts, ask for citations or verification steps.

7. LLM Strengths and Limitations

Strengths	Limitations / risks
Fast drafting and transformation	Can generate confident errors
Natural-language interaction	May misunderstand ambiguous prompts
Summarization of supplied material	Quality depends on input/context
Coding assistance	Generated code still needs testing
Idea generation	Can repeat common patterns or biases

Language tutoring	Not a substitute for authoritative sources in high-stakes domains
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8. AI Tools: A Practical Map

AI tools are applications built around one or more AI models. The model is the engine; the tool adds an interface, workflow, files, search, image generation, audio, video, coding environments or other capabilities.

Category	Typical use	Example output
Text / LLM	Writing, Q&A;, tutoring, analysis	Notes, explanations, drafts
Research / grounded AI	Work with sources and documents	Source-based summaries
Image generation	Create visual concepts	Illustrations, posters
Video generation	Create or transform clips	Short videos
Audio / voice	Speech and sound workflows	Voiceovers, transcripts
Coding AI	Generate, debug, explain code	Programs and fixes
Data / analytics AI	Analyze structured information	Tables, insights, charts

A simple AI content workflow

- 1. Idea → use an LLM to brainstorm.
- 2. Research → collect reliable sources and verify facts.
- 3. Script → use an LLM to structure the content.
- 4. Visuals → use an image/video AI tool.
- 5. Voice → create or record narration.
- 6. Edit → combine visuals, audio, captions and music.
- 7. Review → check facts, copyright, privacy, quality and safety.

9. Google NotebookLM: Detailed Lecture

NotebookLM is Google's AI-powered research and learning tool designed around the sources you provide. Instead of treating a conversation as purely general-purpose Q&A, NotebookLM emphasizes working with a set of uploaded or connected reference materials and generating responses grounded in those materials.

Core idea

Your sources → NotebookLM notebook → AI analysis → grounded answers / study materials

Why this matters

Suppose you have a 70-page PDF, lecture slides, a web source and your own notes. Reading everything manually can be slow. A source-grounded AI notebook can help you ask questions across the material, summarize sections, compare ideas, identify themes and create study-oriented outputs.

Typical source types and materials

- PDFs and documents.
- Web-based source material when supported by the current NotebookLM interface.
- Text and other supported reference materials.
- Lecture notes, study guides, reports and research material.

Important: supported source types, limits and interface features can change over time. Always check the current NotebookLM interface/help documentation for the latest limits and available formats.

10. NotebookLM Workflow

- Step 1 — Create a notebook: Start a notebook for one project, subject or research topic.
- Step 2 — Add sources: Bring in the documents and reference materials you want the AI to work from.
- Step 3 — Ask grounded questions: Ask specific questions about the sources rather than vague questions.
- Step 4 — Inspect citations/references: When the interface provides source citations or source locations, use them to verify the answer.
- Step 5 — Generate study material: Turn source content into summaries, FAQs, study guides, outlines and other supported outputs.
- Step 6 — Iterate: Ask follow-up questions and correct the notebook's direction with clearer instructions.

Example: Engineering student

Imagine you upload your Engineering Chemistry Unit 1 notes, a PDF of lecture slides and a question bank. You could ask: "Using only these sources, explain Molecular Orbital Theory for a first-year student. Then list the important postulates, explain the bond-order formula, and create a 10-mark answer structure."

The key advantage is source grounding: you are asking the system to work with a defined knowledge collection instead of relying only on its general model knowledge.

11. NotebookLM vs a General AI Assistant

Feature	General AI assistant	NotebookLM-style source workflow
Main purpose	Broad conversation and task assistance	Research/learning around supplied sources

Knowledge focus	Model knowledge + available tools/context	Your selected source collection
Best for	Writing, brainstorming, coding, general Q&A;	Notes, PDFs, lectures, source-based study
Verification	May need external checking	Source references can help trace claims
Workflow	Conversation-centered	Notebook/source-centered

12. Retrieval-Augmented Generation (RAG) and NotebookLM

RAG is a general architecture in which a system retrieves relevant information from a knowledge source and gives that information to a generative model as context before generating an answer. Source-grounded notebook workflows are closely related to this idea, although the exact implementation details of a commercial product are not necessarily public.

Simplified RAG flow

- 1. User asks a question.
- 2. System searches the relevant source collection.
- 3. Relevant passages are selected.
- 4. Passages are supplied as context to the language model.
- 5. The model generates an answer grounded in that context.
- 6. Source references can help the user verify the answer.

RAG is especially useful when information is private, specialized, recent or too large to fit comfortably into a normal prompt.

13. Responsible AI Use

- Accuracy: Verify important claims, especially academic, legal, medical, financial and safety-related information.
- Privacy: Do not upload confidential or personal information unless you understand the service's data practices and have permission.
- Copyright: Do not assume that AI generation makes copyrighted material free to reproduce.
- Bias: Models can reflect biases in training data and evaluation processes.
- Academic integrity: Follow your institution's rules about AI-assisted assignments.
- Human judgment: AI should support thinking, not eliminate responsibility for checking important work.

14. Exam-Oriented Questions

Short-answer questions

- What is Artificial Intelligence?
- Define Machine Learning and Deep Learning.
- What is an LLM?
- What is a token in an LLM?
- What is a context window?
- What is prompt engineering?

- What is an AI hallucination?
- What is RAG?
- What is Google NotebookLM?

Long-answer / 10-mark questions

- Explain the architecture and working idea of a modern LLM.
- Explain the stages of LLM training: pre-training, fine-tuning and alignment.
- Explain prompt engineering with suitable examples.
- Discuss the strengths, limitations and responsible use of generative AI.
- Explain Google NotebookLM, its source-grounded workflow, uses for students and advantages over a general-purpose AI assistant.
- Explain RAG and describe why source-grounded AI is useful for research and learning.

15. Quick Revision Sheet

Term	Remember this
AI	Broad field of machines performing tasks associated with intelligence.
ML	Systems learn patterns from data.
Deep Learning	Machine learning using multi-layer neural networks.
Generative AI	AI that creates new content.
LLM	Large neural model for language/token sequences.
Token	A unit used to represent input/output text.
Prompt	Instruction/context supplied to an AI system.
Hallucination	Plausible-sounding but unsupported or incorrect output.
RAG	Retrieve relevant source information, then generate using it as context.
NotebookLM	Source-centered AI research/study workflow around provided materials.

One-line takeaway

LLMs generate language from learned patterns; AI tools wrap models into practical workflows; and NotebookLM is especially useful when you want AI to work closely with a defined collection of your own sources.

Study tip: For a 10-mark answer, use this structure: Definition → working/process → key components → example → advantages → limitations → conclusion.

Prepared as educational lecture notes. Product features and supported formats may change, so current product documentation should be checked for the latest NotebookLM capabilities.